Home delivery option, date and time-slot choices

Transportation and Urban Studio E

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Context

This case study examines how e-commerce customers trade off delivery fees, speed and scheduling flexibility. Customers choose next-day delivery, normal delivery, or scheduled delivery with a specified date and time slot. The data can be used to study willingness to pay for faster delivery and narrower time windows, and the potential for demand management in home delivery. [1, 2]

Data Collection

The stated-choice survey was conducted in Japan from 30 April to 14 May 2021 with Yamato Holdings. Invitations went to Kuroneko Members in the Tokyo, Osaka and Nagoya metropolitan areas. The original survey had 4,872 completed responses, with five tasks per respondent; the cleaned research sample in [1] had 19,860 tasks from 4,062 respondents.

The class file **delivery_option_SP_sample.csv** contains 2,000 sampled choice situations, 66 columns and 1,622 distinct customers (no). Each row records one complete choice, with task attributes in wide format. Customers contribute one to four rows in this subset; retain no for panel estimation and respondent-level train/test splits. The file has no explicit task identifier, so preserve the original row number as a local task key.

Alternatives and choice coding

Choice code	Meaning
`1000`	Next-day delivery: delivery one day after ordering; no selected time slot.
`2000`	Normal delivery: delivery within a stated date range; no selected time slot.
`3000 + 10*d + s`	Scheduled delivery: d = 2,...,8 days after ordering; s = 1 earliest, 2 other/daytime, 3 latest slot. Example: 3051 means day 5, earliest slot.

The model in [1] groups the original time slots into three categories, giving 23 alternatives: two unscheduled options plus 7 days x 3 scheduled slot categories. The "other" category pools intermediate slots; it is not always a single physical time window. The survey delivery period is 09:00-21:00, with slots of 2, 3 or 4 hours.

Use and source scope

Classroom use only. The GroupProject teaching material identifies this dataset as confidential: use it only for this class and remove it from your computer after the group project.

The class sample and supplied codebook are the sources for local column names and counts. Published statistics describe a larger sample and should not be substituted for the statistics of this file.

Variables: task attributes and outcomes

All monetary delivery attributes are in Japanese yen. A suffix d denotes a day after ordering. Names and capitalization match the CSV.

Name	Description / coding
`no`	Customer identifier; repeated across sampled tasks.
`sp_price_next / sp_price_normal / sp_price_specific`	Base delivery fee for next-day / normal / scheduled delivery, respectively. Each takes values 300, 400, 500 or 600 JPY.
`sp_add_weekend`	Additional charge for scheduled delivery on a holiday: 0 or 100 JPY. The paper refers to holidays, not only weekends.
`sp_add_night`	Additional charge for the latest scheduled time slot: 0 or 100 JPY.
`sp_slot_range`	Width of each scheduled time slot: 2, 3 or 4 hours.
`sp_day_after / sp_day_range`	Earliest normal-delivery day (2 or 3 days after ordering) / width of the normal-delivery date range (2, 3 or 4 days).
`norm_day_mean`	Normal-delivery mean time, in days. In all local rows it equals sp_day_after + sp_day_range / 2.
`holiday1 to holiday8`	Holiday indicators for each of the next eight days (1 = holiday, 0 otherwise). The survey period includes Golden Week.
`weekday2 to weekday8`	Day-of-week codes for days 2-8, stored as 0-6. The local codebook does not explicitly define the labels; confirm the convention before attaching weekday names.
`Choice`	Combined delivery-option/date/slot outcome; see the code table on page 1.
`sp_choice`	Original option/date outcome: 100 = next-day; 200 = normal; 302-308 = scheduled delivery on days 2-8.
`sp_choice_slot`	Original time-slot category, 1-13 for scheduled delivery; 1000 when no slot is selected. Exact clock-time labels are not supplied in the local codebook.
`sp_choice_date_after`	Selected day after ordering (2-8); 1000 for next-day/normal choices where the scheduled date question does not apply.
`sp_choice_date_holiday`	Stored holiday flag for the selected date. It disagrees with the corresponding holidayd field in 492 of the 700 scheduled choices; see preparation notes.
`sp_choice_first_slot / sp_choice_last_slot / sp_choice_other_slot`	Indicators of the chosen earliest / latest / intermediate slot. All are zero for next-day and normal delivery.
`sp_slot_choice`	Grouped selected slot: 1 earliest, 2 intermediate, 3 latest; 0 for next-day/normal delivery.

Outcome-derived fields describe the chosen alternative. When constructing utility functions, use the fee, day and slot attributes of every candidate alternative; selected-slot and selected-day flags are not exogenous predictors.

Variables: customer characteristics

Name	Description / coding
gender / male	gender: 1 male, 2 female. male: indicator for gender = 1.
age / generation	Age in years (10-87 in this file) / stored age-decade category (1-8).
age_29 / age_70	Age-group indicators: age <=29 / age >=70.
household / job	Original household-composition / occupation category. Full code lists follow on page 4.
rp_item_category / rp_item_price	Category / price band of the reference online purchase. Full code lists follow on page 4.
ref_item_price / ref_frequency	Preprocessed numeric reference-item price / online-shopping frequency. The local codebook does not document their scaling. Do not interpret these values directly as yen or purchases per week.
member_free	Membership with a free-delivery privilege. The experiment explicitly required payment regardless of membership. [1]
tokyo	Residence in Tokyo indicator.
alone / couple_double	Household indicators: household = 1 (alone) / 2 (couple, both working).
live_child / live_family	Household indicators: household = 4 (with own children) / 6 (with parents, none working, and no own children). The latter is narrower than its column name suggests.
public / self / prof	Occupation indicators: job = 4 (public-sector/teaching/nonprofit) / 6 (self-employed in commerce/services) / 9 (professional).
parttime / nojob	Occupation indicators: job = 10 (part-time) / 13 (not employed).
dep_8 / dep_10 / arr_16 / arr_19	Stored departure/arrival-time indicators. Exact threshold inclusivity and treatment of noncommuters are not specified in the local codebook; confirm before interpreting coefficients.
teleworker / freq_user	Stored teleworking / frequent-online-shopping indicators. Exact frequency cutoffs are not supplied in the local codebook.
house / apparment	Detached-house / apartment indicators. Preserve the original spelling apparment.
box	Presence of a delivery box at home.

Preparation notes

Use the full Choice code for the 23-alternative model, or explicitly aggregate it to three delivery options for a simpler model. The data contain 1,300 structural values of 1000 in the scheduled-date and original-slot fields. These are not delivery times or ordinary missing responses.

For scheduled alternative (d, s), the intended fee is sp_price_specific + sp_add_weekend x holidayd + sp_add_night x I(s = 3). Validate the calendar flags before applying this rule: the selected-date holiday flag is inconsistent with the day-specific flags. Do not silently overwrite the source data.

Original categorical codes

The following labels translate the supplied class codebook. Codes are categories, not continuous quantities. Some documented categories do not occur in the class sample.

Variable	Code labels
`rp_item_category`	1 books/magazines; 2 software/games; 3 computers/peripherals; 4 food/drinks/alcohol; 5 clothing/shoes/accessories; 6 music CDs/video; 7 gifts; 8 kitchenware/household goods; 9 toys; 10 household appliances/AV; 11 health/medical goods; 12 cosmetics/beauty; 13 office supplies; 14 interior goods; 15 other; 16 only large items requiring special shipping; 17 never purchased goods requiring delivery online. The local sample contains codes 1-15.
`rp_item_price`	JPY bands: 1 <1,000; 2 [1,000,3,000); 3 [3,000,5,000); 4 [5,000,7,500); 5 [7,500,10,000); 6 [10,000,15,000); 7 [15,000,20,000); 8 [20,000,30,000); 9 [30,000,50,000); 10 >=50,000; 11 unknown/cannot recall.
`household`	1 living alone; 2 couple/partners, both working; 3 couple/partners, not both working; 4 living with own children; 5 no own children, living with parents, at least one parent working; 6 no own children, living with parents, none working; 7 living with other family (no own children or parents); 8 living with friends/acquaintances; 9 other.
`job`	1 company employee; 2 company manager; 3 business owner/executive; 4 public servant/teacher/nonprofit employee; 5 temporary/contract employee; 6 self-employed in commerce/services; 7 small/home office; 8 agriculture/forestry/fishing; 9 professional (e.g., legal/accounting/medical); 10 part-time worker; 11 homemaker; 12 student; 13 not employed; 14 other.

Descriptive statistics: choices

Delivery option	Count	Share of all tasks
Next-day	856	42.8%
Normal	444	22.2%
Scheduled	700	35.0%
Total	2,000	100.0%

Scheduled slot category	Count	Share of scheduled choices
Earliest	321	45.86%
Intermediate / other	200	28.57%
Latest	179	25.57%
Total	700	100.00%

The first denominator is 2,000 choice tasks; the second is the 700 scheduled choices. Each table contains mutually exclusive outcomes whose shares sum to 100%.

Descriptive statistics: task attributes

Variable	N	Min	Max	Mean	SD
sp_price_next	2,000	300	600	449.15	111.16
sp_price_normal	2,000	300	600	442.30	111.61
sp_price_specific	2,000	300	600	443.85	111.66
sp_add_weekend	2,000	0	100	51.80	49.98
sp_add_night	2,000	0	100	53.65	49.88
sp_slot_range	2,000	2	4	3.00	0.82
sp_day_after	2,000	2	3	2.49	0.50
sp_day_range	2,000	2	4	3.01	0.80
norm_day_mean	2,000	3	5	3.99	0.62

All summaries are computed from the 2,000 local rows. N excludes empty cells; none of the 66 fields has an empty cell in this sample. SD is the sample standard deviation. Customer-level attributes repeated across tasks should be summarized after deduplicating no when the aim is to describe people.

Possible analyses

A three-option multinomial logit provides an initial comparison of delivery services. A richer model uses the 23 coded alternatives to estimate fee, waiting-time and slot-width effects, with shared components for service types or days. Repeated observations permit panel or mixed-logit specifications. State clearly whether the intermediate-slot category represents an aggregate of several original slots.

References and sources

[1] Oyama, Y., Fukuda, D., Imura, N. and Nishinari, K. (2024). Do people really want fast and precisely scheduled delivery? E-commerce customers' valuations of home delivery timing. Journal of Retailing and Consumer Services, 78, 103711. Local copy: delivery_option_SP_paper1.pdf.
https://doi.org/10.1016/j.jretconser.2024.103711

[2] Okazaki, R., Oyama, Y., Imura, N. and Nishinari, K. (2025). Evaluating choice-based demand management strategies for day-to-day home delivery planning. Research in Transportation Economics, 113, 101615. Local copy: delivery_option_SP_paper2.pdf.
https://doi.org/10.1016/j.retrec.2025.101615

Source pages checked on 10 November 2026. Counts and descriptive statistics refer to the local CSV files, excluding headers. This teaching description was prepared from the cited sources and local file inspection; it is not an author-supplied codebook.

Local teaching sources: GroupProject.pdf / GroupProject.pptx, delivery_option_SP_description.txt, and delivery_option_SP_sample.csv.